

AI TRAINING – DAY 2 ASSIGNMENT

Python Fundamentals: Setup → Variables → Data Types → Input/Output → Type Conversion

Practical + MCQ + Debugging + Mini Challenge

Level	Beginner → Medium
Suggested Time	3–4 hours
Primary Goal	Understand Python basics by writing and fixing programs
Programs	10 required programs + debugging + challenge
Submission	One document/PDF + .py files or screenshots/output

1. Learning Goals

- Set up Python and verify it works.
- Understand variables and Python values.
- Understand int, float, str, bool and None.
- Understand input() and print().
- Understand why input() normally returns a string.
- Convert values between common data types.
- Write small programs independently.
- Read and fix simple Python errors.
- Explain every program in your own words.

2. Rules for This Assignment

- Type every program yourself; do not copy complete solutions.
- You may use AI to explain an error after attempting it yourself.
- For every program write: Input → Processing → Output.
- Test every program with at least 2 different inputs.
- Keep track of errors you faced and what you changed.
- Use Python 3.

3. Part A – Python Setup

1. Install Python 3 or use an online Python environment.
2. Verify the Python version from the terminal/command prompt.
3. Create a folder named Day02_Python_Basics.
4. Create hello.py and print: Hello, Python!
5. Run the file successfully.
6. Take a screenshot showing the version and successful output.

Expected evidence: Python version + hello.py + output.

4. Part B – Explore Before Coding

1. What does `type(variable)` tell you?

Ans: it tells us what type of data is being stored in variable.

2. Create an int, float, str, bool and None value. Check their types.

```
a=10
b=1.7
c="deva"
d=True
e=None
print(type(a))
print(type(b))
print(type(c))
print(type(d))
print(type(e))
```

3. What happens when you add an int and a float?

Ans: Both values will be added and then the final output will be float data type.

4. What happens when you join two strings with `+`?

Ans: both strings will be added and final output will be a string without spaces.

Ex: `print("deva"+"ntr")`

Output: devantr

5. What type does `input()` return when the user enters 25?

Ans: type of 25 is string.

6. What happens with `10 + input('Enter number: ')`? Why?

It will be unsupported operation because 10 is integer and `input()` value is default datatype is string.

7. What is the difference between 10 and '10'?

Ans: 10 is integer and '10' is a string.

8. What is the difference between True and 'True'?

Ans: True Boolean type and 'True' is a string.

9. What does `int('25')` produce?

Ans: at here '25' is converted to integer.

10. What does `int('25.5')` do? Test it.

Ans: it is error because int can directly convert str to int with whole number not decimal number.

11. What does str(25) produce?

Ans: at initial 25 is integer after that it converts into string.

12. What do bool(0), bool(1), bool("") and bool('hello') produce?

```
print(bool(0))#False
print(bool(1))#True
print(bool(""))#False
print(bool('hello'))#True
```

13. What does None mean? Give one practical example.

Ans: None means no value or data in python.

Ex: student_name=None

At the above variable name is created but no value has been assigned.

5. Part C – Concept Understanding

Concept	Explain in your own words
Variable	Variable is name that is used to store values. so we use that variable later. variables store the different types of data types. Ex: number=0 At the above number is variable. 0 is the value.
int	stores the integer values ex: 1,2,3,4,55,6,7,.....
float	it stores the float values ex: 1.2, 3.6, 9.666,.....
str	String is used to store the alphabetical words. Ex: "apple", "deva".....
bool	it has only two values TRUE or FALSE. Ex: print(bool(1)) #True
None	None means no value or no data in Python. It represents the absence of a value. Ex: student_name=None
input()	It takes input from the user through keyboard. Num=input("enter number:")
print()	It prints the information on the screen. Print("Hello world")
Type conversion	Type conversion is nothing converting one data type into another data type. Num=5 Print(str(Num))

6. Part D – MCQs: Basic Level

1. Which function displays output?

- A. input()
- B. print()
- C. output()
- D. show()

2. Which stores whole numbers?

- A. float
- B. str
- C. int
- D. bool

3. What is 10.5?

- A. int
- B. float
- C. str
- D. bool

4. Which is a string?

- A. 25
- B. 25.0
- C. True
- D. '25'

5. Which is a boolean?

- A. yes
- B. TRUE
- C. True
- D. 'True'

6. What is type(None)?

- A. null
- B. NoneType
- C. empty
- D. bool

7. Which gets user input?

- A. scan()
- B. read()
- C. input()
- D. get()

8. What does str(100) produce?

- A. int 100

- B. 100.0
- C. '100'
- D. True

9. What does `int('20')` produce?

- A. '20'
- B. 20
- C. 20.0
- D. Always error

10. Which is a valid variable name?

- A. 2name
- B. student-name
- C. student_name
- D. class

7. Part E – MCQs: Medium Level

1. What type is `input('Age: ')`?

- A. int
- B. float
- C. str
- D. bool

2. What happens in `10 + '5'`?

- A. 15
- B. 105
- C. TypeError
- D. 10.5

3. What is `int(12.9)`?

- A. 13
- B. 12
- C. 12.9
- D. Error

4. What is `float('10')`?

- A. 10
- B. '10.0'
- C. 10.0
- D. Error

5. What is `str(True)`?

- A. True
- B. 'True'

- C. 1
- D. '1'

6. What is bool(0)?

- A. True
- B. False
- C. 0
- D. Error

7. What is 5 + 2.5?

- A. 7
- B. 7.5
- C. '7.5'
- D. Error

8. Correct integer input?

- A. int(input())
- B. input(int())
- C. integer(input())
- D. input().int()

9. What is x after x=None?

- A. 0
- B. False
- C. "
- D. None

10. Which can fail if user enters abc?

- A. int(input())
- B. str(input())
- C. input()
- D. print(input())

8. Part F – MCQs: Thinking Level

1. Why does age=input() not work with age+5?

- A. input is slow
- B. input returns str
- C. + is unsupported
- D. age becomes None

2. Type conversion means:

- A. Changing a variable name
- B. Changing a value's data type
- C. Deleting a variable

- D. Printing a value

3. What is `int('10.5')`?

- A. 10
- B. 10.5
- C. ValueError
- D. TypeError

4. Which produces 25.0?

- A. `float('25')`
- B. `int('25')`
- C. `str(25)`
- D. `bool(25)`

5. None represents:

- A. zero
- B. False
- C. absence of a value
- D. empty string

6. `x='5', y=2`. Which gives 52?

- A. `x+y`
- B. `int(x)+y`
- C. `x+str(y)`
- D. `float(x)+y`

7. Marks for an average should normally be:

- A. str
- B. bool
- C. int or float
- D. None

8. `price=input(); total=price*2` means:

- A. always numeric
- B. string multiplication may repeat text
- C. price becomes float
- D. Python converts automatically

9. Best first step for a `TypeError`?

- A. Delete code
- B. Ignore it
- C. Read the error and inspect types
- D. Restart PC

10. Which is true?

- A. Variables have fixed types forever
- B. Variables can refer to values of different types
- C. Python has no types
- D. input is automatically int

9. Part G – 10 Required Practical Programs

Program	Task	Topics	Level
1 – Personal Introduction	Ask name, age and city. Print a simple introduction.	str + int + input + print	Basic
2 – Basic Calculator	Ask two numbers. Print sum, difference, product and division.	int/float + arithmetic	Basic
3 – Shopping Bill	Ask item name, quantity and price. Calculate total.	str + int + float + conversion	Basic
4 – Temperature Converter	Ask Celsius and convert to Fahrenheit.	float + arithmetic	Basic
5 – Student Marks	Ask 3 subject marks. Calculate total and average.	numbers + conversion	Basic → Medium
6 – Age Calculator	Ask age and show age after 5, 10 and 20 years.	int + arithmetic	Medium
7 – Type Inspector	Show what input() returns, then convert the value to int/float where possible.	str + type + conversion	Medium
8 – Bill Splitter	Ask bill, number of people and tip %. Calculate total and per-person amount.	float + int + conversion	Medium
9 – Data Type Challenge	Create int, float, str, bool and None. Print values/types. Convert selected values and explain.	All required types	Medium
10 – Mini Profile Generator	Ask name, age, height, student status and optional phone. Display values and types.	All topics	Medium

For EVERY program submit:

Problem statement _____

Input _____

Processing / formula _____

Output _____

Your code _____

Test case 1 _____

Test case 2 _____

Error faced and fix _____

One thing learned _____

Programs :

1 – Personal Introduction

Problem statement:

Create a program that asks the name, age, and city and displays personal introduction.

Input:

Name, age, and city.

Processing / formula:

it takes the name and city as string and age as int

Output:

Display the user's personal introduction.

Code:

```
name=input("enter your name:")
age=int(input("enter your age :"))
city=input("enter your city name :")
print("my name is",name)
print(f"I am {age} years old")
print(f"I live in {city}")
```

Test case 1:

Input: Devendra, 21, Hyderabad

Output: My name is Devendra, I am 21 years old, I live in Hyderabad.

Test case 2:

Input: Rahul, 20, Vijayawada

Output: My name is Rahul, I am 20 years old, I live in Vijayawada.

Error faced and fix:

the data type of age is int. if we gave string it gives the type error.

One thing learned:

input() always returns the string.

2 – Basic Calculator

Problem statement:

Create a calculator that asks for two numbers and calculates addition, subtraction, multiplication, division.

Input:

Two numbers.

Processing / formula:

- Sum = num1 + num2
- Difference = num1 - num2
- Product = num1 * num2
- Division = num1 / num2

Output:

Sum, difference, product, and division.

Code :

```
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))
print("Sum:", num1 + num2)
print("Difference:", num1 - num2)
print("Product:", num1 * num2)
print("Division:", num1 / num2)
```

Test case 1:

Input: 10, 5

Output: 15.0, 5.0, 50.0, 2.0

Test case 2:

Input: 20, 4

Output: 24.0, 16.0, 80.0, 5.0

Error faced and fix:

If the second number is 0, division causes ZeroDivisionError.

One thing learned:

by giving int as input it may error when user enter float number.

3 – Shopping Bill

Problem statement:

Create a program that asks for an item name, quantity, price, then calculates the total bill.

Input:

Item name, quantity, and price.

Processing / formula:

Total = quantity × price

Output:

Item name and total bill.

Your code:

```
item = input("Enter item name: ")
quantity = int(input("Enter quantity: "))
price = float(input("Enter price: "))
```

```
total = quantity * price
print("Item:", item)
print("Total:", total)
```

Test case 1:

Input: Pen, 5, 10

Output: Item: Pen, Total: 50.0

Test case 2:

Input: Book, 3, 120

Output: Item: Book, Total: 360.0

Error faced and fix:

If quantity is entered as str(two), int() gives ValueError. Enter a numeric quantity.

One thing learned:

Different data types can be used together in one program.

4 – Temperature Converter

Problem statement:

Ask for the temperature in Celsius and convert it to Fahrenheit.

Input:

Temperature in Celsius.

Processing / formula:

Fahrenheit = (Celsius $\times$ 9/5) + 32

Output:

Temperature in Fahrenheit.

Your code:

```
celsius = float(input("Enter temperature in Celsius: "))
fahrenheit = (celsius * 9 / 5) + 32
print("Temperature in Fahrenheit:", fahrenheit)
```

Test case 1:

Input: 0

Output: 32.0

Test case 2:

Input: 100

Output: 212.0

Error faced and fix:

no error occurred.

One thing learned:

We can use formulas in python

5 – Student Marks

Problem statement:

Create a marks calculator take input as marks of three subjects

Input:

Three subject marks.

Processing / formula:

Total = mark1 + mark2 + mark3

Average = Total / 3

Output:

Total and average marks.

Code:

```
mark1 = float(input("Enter marks for subject 1: "))
mark2 = float(input("Enter marks for subject 2: "))
mark3 = float(input("Enter marks for subject 3: "))
total = mark1 + mark2 + mark3
average = total / 3
print("Total marks:", total)
print("Average marks:", average)
```

Test case 1:

Input: 80, 70, 90

Output: Total: 240.0

Average: 80.0

Test case 2:

Input: 75, 85, 95

Output: Total: 255.0

Average: 85.0

Error faced and fix:

No errors occurred.

One thing learned:

average can be found by addition of observations by number of observations.

6 – Age Calculator

Problem statement:

Ask for the current age and show the age after 5, 10, and 20 years.

Input:

Current age.

Processing / formula:

- After 5 years = age + 5

- After 10 years = age + 10

Output:

Future ages.

Your code:

```
age = int(input("Enter your current age: "))
print("Age after 5 years:", age + 5)
print("Age after 10 years:", age + 10)
print("Age after 20 years:", age + 20)
```

Test case 1:

Input: 20

Output: 25, 30.

Test case 2:

Input: 25

Output: 30, 35.

Error faced and fix:

If age is entered as string, int() gives an error.

One thing learned:

we can add numbers after giving as input.

7 – Type Inspector

Problem statement:

Take a value from the user, show its data type, convert it into another data type.

Input:

one value

Processing / formula:

Check the input type and convert it to int or float.

Output:

number and their type.

Code:

```
number=input("enter number:")
print(number)
print(type(number))
number=int(number)
print(type(number))
print(number)
number=float(number)
print(type(number))
print(number)
```

Test case 1:

Input: 25

Output: str → int

Test case 2:

Input: 10.5

Output: str → float

Error faced and fix:

in type conversion int("10.5") cannot convert into integer.

One thing learned:

input() always give string.

8 – Bill Splitter

Problem statement:

Ask for the bill amount, number of people, and tip percentage. Find the total bill and amount for each person.

Input:

Bill, number of people, and tip percentage.

Processing / formula:

Tip = Bill × Tip% / 100

Total = Bill + Tip

Per person = Total / People

Output:

Tip, total bill, and amount per person.

Your code:

```
bill = float(input("Enter bill amount: "))
people = int(input("Enter number of people: "))
tip_percent = float(input("Enter tip percentage: "))
tip = bill * tip_percent / 100
total = bill + tip
per_person = total / people
print("Tip:", tip)
print("Total bill:", total)
print("Amount per person:", per_person)
```

Test case 1:

Input: 1000, 4, 10

Output: Tip: 100

Total: 1100

Per person: 275

Test case 2:

Input: 500, 2, 5

Output: Tip: 25

Total: 525

Per person: 262.5

Error faced and fix:

If people is 0, division gives an error.

One thing learned:

we can solve realtime problems with python.

9 – Data Type Challenge

Problem statement:

Create different data types and print their values and types.

Input:

No user input.

Processing / formula:

Create int, float, str, bool, and None values and check their types.

Output:

Values and their data types.

Your code:

```
number = 25
price = 10.5
name = "Devendra"
is_student = True
phone = None
print(number, type(number))
print(price, type(price))
print(name, type(name))
print(is_student, type(is_student))
print(phone, type(phone))
number=str(number)
print(number,type(number))
is_student=int(is_student)
print(is_student,type(is_student))
```

Test case 1:

25 → int

10.5 → float

Test case 2:

"Devendra" → str

True → bool

Error faced and fix:

Invalid values cannot always be converted to another type.

One thing learned:

bool can be converted into integer.

10 – Mini Profile Generator

Problem statement:

Ask for personal details and display the values with their data types.

Input:

Name, age, height, student status, and phone number.

Processing / formula:

Convert age to int, height to float, and student status to bool.

Output:

Profile details and their types.

Your code:

```
name = input("Enter your name: ")
age = int(input("Enter your age: "))
height = float(input("Enter your height: "))
student_input = input("Are you a student? (yes/no): ")
is_student = student_input.lower() == "yes"
phone = input("Enter your phone number (optional): ")
print("\n--- Profile ---")
print("Name:", name, type(name))
print("Age:", age, type(age))
print("Height:", height, type(height))
print("Student:", is_student, type(is_student))
print("Phone:", phone, type(phone))
```

Test case 1:

Input: Devendra, 21, 5.8, yes, 9876543210

Output: Name, age, height, student status, and phone details.

Test case 2:

Input: Rahul, 22, 6.0, no

Output: Profile details with student status as False.

Error faced and fix:

If age or height is not a number, conversion gives an error.

One thing learned:

Can used to generate profiles.

10. Part H – Debugging Lab

Bug 1

```
age=input('Enter age: ')
print(age+5)
```

Explain and fix.

Corrected code / explanation: it is type error because age is string and 5 is integer.

```
age=int(input('Enter age: '))
print(age+5)
```

Bug 2

```
price=input('Price: ')
quantity=input('Quantity: ')
print(price*quantity)
```

Explain and fix.

Corrected code / explanation: both inputs are strings. we cannot multiply.

```
price=int(input('Price: '))
quantity=int(input('Quantity: '))
print(price*quantity)
```

Bug 3

```
marks='90'
print(marks+10)
```

Explain and fix.

Corrected code / explanation: marks is string but we are adding integer to it.

```
marks=90
print(marks+10)
```

Bug 4

```
temperature=float(input('Temperature: '))
print('Temperature is '+temperature)
```

Explain and fix.

Corrected code / explanation: temperature is float we cannot add to string .

```
temperature=float(input('Temperature: '))
print('Temperature is ',temperature)
```

Bug 5

```
number=int(input('Number: '))
print('Number: '+number)
```

Explain and fix.

Corrected code / explanation: number is not added to string.

```
number=int(input('Number: '))
print('Number: ',number)
```

Bug 6

```
value=int(input('Number: '))  
print(float(value))
```

Does this work? Explain why.

Corrected code / explanation: there is no error it will work.

11. Part I – Medium Challenge: Student Result Calculator

- Ask student name.
- Ask marks in 5 subjects and convert them to numbers.
- Calculate total and average.
- Display name, total and average.
- Create result_status as a boolean showing the calculation is completed.
- Create remarks=None first, then assign a suitable message.
- Display the type of important variables.
- Test with at least 3 different mark sets.
- Bonus: support decimal marks using float.

```
name = input("Enter student name: ")  
mark1 = float(input("Enter marks for subject 1: "))  
mark2 = float(input("Enter marks for subject 2: "))  
mark3 = float(input("Enter marks for subject 3: "))  
mark4 = float(input("Enter marks for subject 4: "))  
mark5 = float(input("Enter marks for subject 5: "))  
total = mark1 + mark2 + mark3 + mark4 + mark5  
average = total / 5  
result_status = True  
remarks = None  
if average >= 40:  
    remarks = "Pass"  
else:  
    remarks = "Fail"  
print("\n--- Student Result ---")  
print("Name:", name)  
print("Total:", total)  
print("Average:", average)  
print("Result status:", result_status)  
print("Remarks:", remarks)  
print("\n--- Data Types ---")  
print("Name type:", type(name))  
print("Mark type:", type(mark1))  
print("Total type:", type(total))  
print("Average type:", type(average))  
print("Result status type:", type(result_status))  
print("Remarks type:", type(remarks))
```

12. Part J – Programmer Thinking

Why doesn't Python automatically convert '25' to 25?

Answer: because they both are from different datatypes.

When would you choose int instead of float?

Answer: when I need both float and integer values as input.

Why is 'False' different from False?

Answer: 'False' is a string. False is Boolean value.

Why is None useful?

Answer: no need to give value.

What is the difference between a syntax error and a type-related error?

Answer: syntax error comes when the code structure brakes the python rules. type related error occurred when an operation performed on two different data types which did not work.

Which concept was easiest? Why?

Answer: variables and typeconversion

Which was confusing and what experiment helped?

Answer: all topics are understand easily.

13. Self-Check

Skill	Yes	Need Practice	Notes
Run a Python file	☒	☐	
Create variables	☒	☐	
Identify int/float/str/bool/None	☒	☐	
Explain input()	☒	☐	
Use print()	☒	☐	
Perform type conversion	☒	☐	
Understand bool conversion	☒	☐	
Read a simple error	☒	☐	
Write a small program independently	☒	☐	
Explain your own code	☒	☐	

14. Submission Checklist

- Setup screenshot
- Exploration answers
- Concept table
- 30 MCQ answers
- 10 programs + 2 tests each

- Debugging corrections
- Medium challenge
- Reflection answers
- Self-check

15. Suggested Evaluation

Area	What is evaluated	Weight
Concepts + MCQs	Basic understanding	20%
10 Programs	Logic, input/output, conversion	35%
Debugging	Error identification and correction	15%
Medium Challenge	Combining concepts	20%
Explanation	Can explain own code	10%

Day 2 principle: Don't memorize Python. Run it, break it, understand the error, fix it, and explain why it works.